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Rudi Smith

Curriculum Vitae

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https://smithrgh.github.io | scholar.google.com/citations?user=IiKelN0AAAAJ

Education

Virginia Tech Aug. 10, 2024 – Present
Ph.D. Candidate in Applied Mathematics, supervised by S. W. R. Werner and A. Międlar. *Blacksburg, VA*

University of Oxford Oct. 13, 2019 – Jun. 28, 2024
Master of Mathematics, supervised by Y. Nakatsukasa. *Exeter College, Oxford*

Awards

East Scholarship 2020 – 2024
Academic scholarship awarded based on sustained high performance. *Exeter College, Oxford*

Patrick Prize in Mathematics 2021
Awarded for outstanding academic performance in the second year of study. *Exeter College, Oxford*

Potter Prize in Mathematics 2020
Awarded for outstanding academic performance in the first year of study. *Exeter College, Oxford*

Experience & Current Research

Doctoral Research Aug. 10, 2024 – Present
Randomized tensor methods and numerical linear algebra for model reduction. *Blacksburg, VA*

Graduate Research Assistant Aug. 10, 2026 – Present
Acceleration methods for tensors. *Blacksburg, VA*

Computing Scholar, Lawrence Livermore National Laboratory Jun. 1, 2026 – Aug. 21, 2026
Fast multi-linear transforms and tensor compression, supervised by P. Lindstrom and A. Fox. *Livermore, CA*

Graduate Research Assistant Jan. 10, 2026 – May 9, 2026
Acceleration methods for model reduction. *Blacksburg, VA*

Graduate Research Assistant Aug. 10, 2025 – Dec. 24, 2025
Tangential methods for matrix equations. *Blacksburg, VA*

Master's Dissertation Apr. 22, 2024
'An Exploration of Trace Estimation Algorithms with an Emphasis on Exact Recovery'. *Exeter College, Oxford*

Undergraduate Research Internship (Funded by AMD) Jun. 26, 2023 – Aug. 4, 2023
Randomized initialization and pruning of neural networks, supervised by J. Tanner. *Exeter College, Oxford*

Teaching

Teaching Certification Aug. 26, 2024 – May 5, 2025
Completed practical training and gained formal eligibility to teach at Virginia Tech. *Blacksburg, VA*

Graduate Teaching Assistant Aug. 10, 2024 – Jul. 7, 2025
Graduate teaching assistant for undergraduate mathematics courses: *Blacksburg, VA*

- CMDA 3605: Mathematical Modeling: Methods and Tools I.
- CMDA 3606: Mathematical Modeling: Methods and Tools II.
- MATH 1025: Elementary Calculus.

Mathematics Tutor Aug. 14, 2021 – Aug. 5, 2024
Tutor for university admission tests and competitive mathematics (STEP, MAT, AMC12, AIME). *Exeter College, Oxford*

Service & Leadership

Vice President SIAM Student Chapter — Aug. 10, 2026 – Present
Organize and run events for students interested in applied mathematics. — *Blacksburg, VA*

Graduate Peer Mentoring Program Mentor — Aug. 21, 2025 – Present
Mentor for first-year graduate students. — *Blacksburg, VA*

Directed Reading Program Mentor — Sep. 7, 2026 – Dec. 9, 2026
Mentor for undergraduate students studying machine learning and tensors. — *Blacksburg, VA*

Math Circle Presenter — Oct. 4, 2025 and Oct. 24, 2026
Tuition-free enrichment program open to all students in grades 4–8. — *Blacksburg, VA*

Reading Group Organizer — Aug. 25, 2025 – Dec. 10, 2025
Led a graduate-level reading group on high-dimensional probability. — *Blacksburg, VA*

Conference Contributions

Oral Presentations

SIAM Applied Linear Algebra (SIAM LA) — May 24, 2027 – May 28, 2027
(Invited minisymposium talk) Title TBD — *Long Beach, CA*

Joint Mathematics Meetings, AMS — Jan. 12, 2027 – Jan. 15, 2027
(Invited minisymposium talk) 'Randomized Tucker-Sketched GMRES' — *Chicago, IL*

SIAM Mathematics of Data Science (SIAM MDS) — Nov. 16, 2026 – Nov. 20, 2026
(Invited minisymposium talk) 'Randomized Techniques for Tucker GMRES' — *Salt Lake City, UT*

Applied Numerical Analysis Seminar — Nov. 2, 2026
(Invited seminar talk) 'Randomized Tucker-sketched GMRES' — *Blacksburg, VA*

Young Mathematicians in Model Order Reduction (YMMOR) — May 25, 2026 – May 29, 2026
(Talk) 'A Tangential Low-rank ADI Method for Solving Indefinite Lyapunov Equations' — *Blacksburg, VA*

International Linear Algebra Society (ILAS) — May 18, 2026 – May 22, 2026
(Invited minisymposium talk) 'A Tangential Low-rank ADI Method for Solving Indefinite Lyapunov Equations' — *Blacksburg, VA*

ICERM: Stochastic and Randomized Algorithms in Scientific Computing — Jan. 20, 2026 – Apr. 24, 2026
(Seminar) 'A Tangential Low-rank ADI Method for Solving Indefinite Lyapunov Equations' — *Providence, RI*

Applied Numerical Analysis Seminar — Oct. 27, 2025
(Invited seminar talk) 'A Tangential ADI Method for the Efficient Solution of Large-scale Indefinite Lyapunov Equations' — *Blacksburg, VA*

Applied Numerical Analysis Seminar — Sep. 30, 2024
(Invited seminar talk) 'Exploiting Sparsity in Trace Estimation' — *Blacksburg, VA*

Presented Posters

IMSI: Reduced Order and Surrogate Modeling for Digital Twins — Nov. 10, 2025 – Nov. 14, 2025
'Tangential Quadrature-based Balanced Truncation' — *Chicago, IL*

IMSI: Data Assimilation and Inverse Problems for Digital Twins — Oct. 6, 2025 – Oct. 10, 2025
'Structured Sketching for Fast and Scalable Tucker Tensor Summation' — *Chicago, IL*

IMSI: Kernel Methods in Uncertainty Quantification and Experimental Design — Mar. 31, 2025 – Apr. 4, 2025
'Efficient Sketching Based Summation of Tucker Tensors' — *Chicago, IL*

Workshops

ICERM: Asynchronous Methods for Numerical Linear Algebra May 4, 2026 – May 8, 2026
Workshop Attendee. *Providence, RI*

ICERM: Stochastic and Randomized Algorithms in Scientific Computing Jan. 20, 2026 – Apr. 24, 2026
Long Program Attendee. *Providence, RI*

IPAM: Research Collaboration Workshop Aug. 11, 2025 – Aug. 15, 2025
Topic: 'Randomization in Transformer Models'. *Los Angeles, CA*

Publications

Journal Articles

[1] R. Smith, M. Pasha, A. Galindo-Olarte, H. Al Daas, G. Ballard, J. Nakao, J.-M. Qiu, and W. Taitano. (2026). "Efficient sketching-based summation of Tucker tensors." *Accepted to SIAM Journal on Scientific Computing (SISC)*. arXiv: 2603.13532. doi: 10.48550/arXiv.2603.13532

Submitted

[1] M. F. Serret, A. Cortinovis, Y. Dong, D. Halikias, A. Ma, F. Matti, D. Needell, K. J. Pearce, E. Rebrova, D. Shur, R. Smith, H.-X. Wang, and L. Grigori. (2026). "Attention mechanisms through the lens of numerical methods: Approximation methods and alternative formulations." *Submitted*. arXiv: 2604.01757. doi: 10.48550/arXiv.2604.01757

[2] R. Smith and S. W. R. Werner. (2025). "A tangential low-rank ADI method for solving indefinite Lyapunov equations." *Submitted*. arXiv: 2512.04983. doi: 10.48550/arXiv.2512.04983

Preprint

[1] A. Bucci, M. Iannacito, M. Pasha, and R. Smith. (2026). "Randomized Tucker-sketched GMRES." *Preprint*. arXiv: 2608.11091. doi: 10.48550/arXiv.2608.11091